

DAVIS-MONTHAN AFB, AZ, USA

WMO: 722745

Lat: 32.167N

Lon: 110.883W

Elev: 2704

StdP: 13.32

Time Zone: -7.00 (W07)

Period: 94-19

WBAN: 23109

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	32.1	35.1	0.1	6.1	63.5	4.6	7.7	65.1	23.7	61.2	20.4	57.9	5.5	150	0.463

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.2	105.3	65.3	103.1	65.1	100.8	64.8	72.9	85.3	72.0	85.8	71.2	86.1	11.1	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
70.3	123.3	76.1	68.6	116.4	76.7	67.4	111.5	77.5	38.2	84.6	37.4	85.2	36.6	85.7	79.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.7	18.4	16.3	DB	26.1	109.3	3.3	2.8	23.7	111.2	21.8	112.9	19.9	114.4	17.5	116.4	
			WB	22.3	75.3	3.7	1.8	19.7	76.6	17.6	77.6	15.5	78.6	12.9	79.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	70.1	53.5	55.6	61.8	67.2	75.7	85.6	87.3	86.1	82.1	71.8	61.5	52.7
	DBStd	14.01	6.66	6.78	6.71	7.20	6.35	4.84	4.18	3.81	4.24	6.78	7.34	6.70
	HDD50	131	42	24	5	1	0	0	0	0	0	8	51	
	HDD65	1385	359	266	140	60	6	0	0	0	24	147	383	
	CDD50	7485	150	182	369	518	796	1068	1156	1120	963	675	354	134
	CDD65	3262	1	4	39	126	337	618	691	655	513	234	43	1
	CDH74	45403	54	148	785	1958	4902	9709	9667	8666	6268	2641	571	34
	CDH80	25857	4	16	198	798	2688	6351	5914	5095	3466	1184	142	1
Wind	WSAvg	7.6	7.0	7.2	7.7	8.3	8.1	8.1	7.9	7.4	7.4	7.5	7.3	6.9
Precipitation	PrecAvg	11.5	0.9	0.8	0.6	0.3	0.2	0.2	2.3	2.1	1.5	0.9	0.6	1.0
	PrecMax	21.9	4.8	3.3	2.2	1.3	1.1	1.6	6.8	4.9	5.8	5.0	2.2	5.0
	PrecMin	6.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.2	0.0	0.0	0.0	0.0
	PrecStd	3.1	0.9	0.8	0.6	0.3	0.3	0.3	1.5	1.2	1.3	1.2	0.6	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	79.1	82.3	88.7	95.0	102.2	108.5	108.4	106.2	102.1	96.6	88.2	77.5
		MCWB	51.4	52.1	54.7	57.3	60.4	64.6	66.4	68.5	65.8	61.0	56.2	51.3
	2%	DB	74.9	78.7	84.7	91.0	99.0	105.5	105.0	103.2	99.5	93.1	84.0	73.8
		MCWB	50.1	51.2	53.7	56.0	59.3	63.4	66.4	68.0	64.5	60.5	55.0	50.7
	5%	DB	71.5	74.9	81.8	88.2	95.6	103.5	102.3	100.4	97.3	90.3	80.8	70.3
		MCWB	49.1	50.5	52.6	55.0	58.2	62.9	66.9	67.9	65.0	59.8	54.3	49.8
	10%	DB	67.6	70.7	78.8	85.0	92.9	100.9	99.8	98.1	94.9	87.1	76.9	66.2
		MCWB	48.0	49.3	51.6	53.9	57.2	62.2	66.8	68.2	65.1	58.9	53.4	48.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	56.9	57.5	57.7	59.7	63.5	72.5	74.2	74.2	72.5	68.9	60.8	56.8
		MCDB	63.8	65.7	77.8	87.6	90.1	85.1	83.9	85.0	84.4	76.7	70.5	63.0
	2%	WB	53.8	55.1	55.7	57.7	61.4	69.5	72.8	73.0	71.2	66.1	58.4	53.9
		MCDB	65.0	67.2	76.5	85.8	90.5	88.3	84.7	85.7	83.3	80.5	74.7	67.0
	5%	WB	51.7	53.3	54.2	56.1	59.9	67.7	71.9	72.1	70.2	63.7	56.6	51.8
		MCDB	65.1	66.6	76.1	83.7	90.1	90.9	85.4	86.6	83.2	81.7	74.8	65.8
	10%	WB	49.9	51.3	52.7	54.6	58.5	65.7	71.1	71.3	69.2	61.8	54.8	50.0
		MCDB	64.2	66.9	75.1	81.5	89.2	93.9	85.8	87.3	83.3	80.8	73.6	63.6
Mean Daily Temperature Range	5% DB	MDBR	24.4	24.9	27.6	28.8	29.5	29.2	23.2	23.2	23.7	26.0	25.3	23.6
		MCDBR	29.5	31.2	32.0	32.5	32.4	30.8	27.0	26.4	26.7	29.4	29.4	27.9
		MCWBR	13.8	13.9	12.9	13.0	12.0	10.8	7.1	6.5	7.8	10.7	12.5	13.3
	5% WB	MCDBR	23.3	24.1	28.1	29.7	29.6	24.7	21.8	22.0	20.7	23.0	24.1	23.6
		MCWBR	11.8	11.4	11.5	12.0	10.8	8.2	5.3	5.7	6.1	8.9	10.9	12.0
Clear-Sky Solar Irradiance	taub	0.262	0.266	0.287	0.297	0.309	0.332	0.391	0.378	0.336	0.299	0.272	0.261	
	taud	2.634	2.628	2.536	2.495	2.499	2.439	2.406	2.431	2.519	2.567	2.601	2.628	
	Ebn at Noon	312	319	317	315	310	300	283	286	296	302	305	306	
	Edh at Noon	24	26	31	34	34	36	37	36	32	28	24	23	
All-Sky Solar Radiation	RadAvg	1141	1439	1912	2350	2600	2601	2161	2046	1877	1603	1263	1044	
	RadStd	81	116	95	74	96	138	134	74	90	101	62	49	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (11 neighbors)	+0.49	N/A	N/A	N/A	N/A	N/A	N/A	-131	+191	N/A

Nomenclature: See separate page